



Jazz Pre-Configured Processor Cores

Products & Solution Overview





PIONEERING A **NEW GENERATION** OF IP

The Improv Advantage

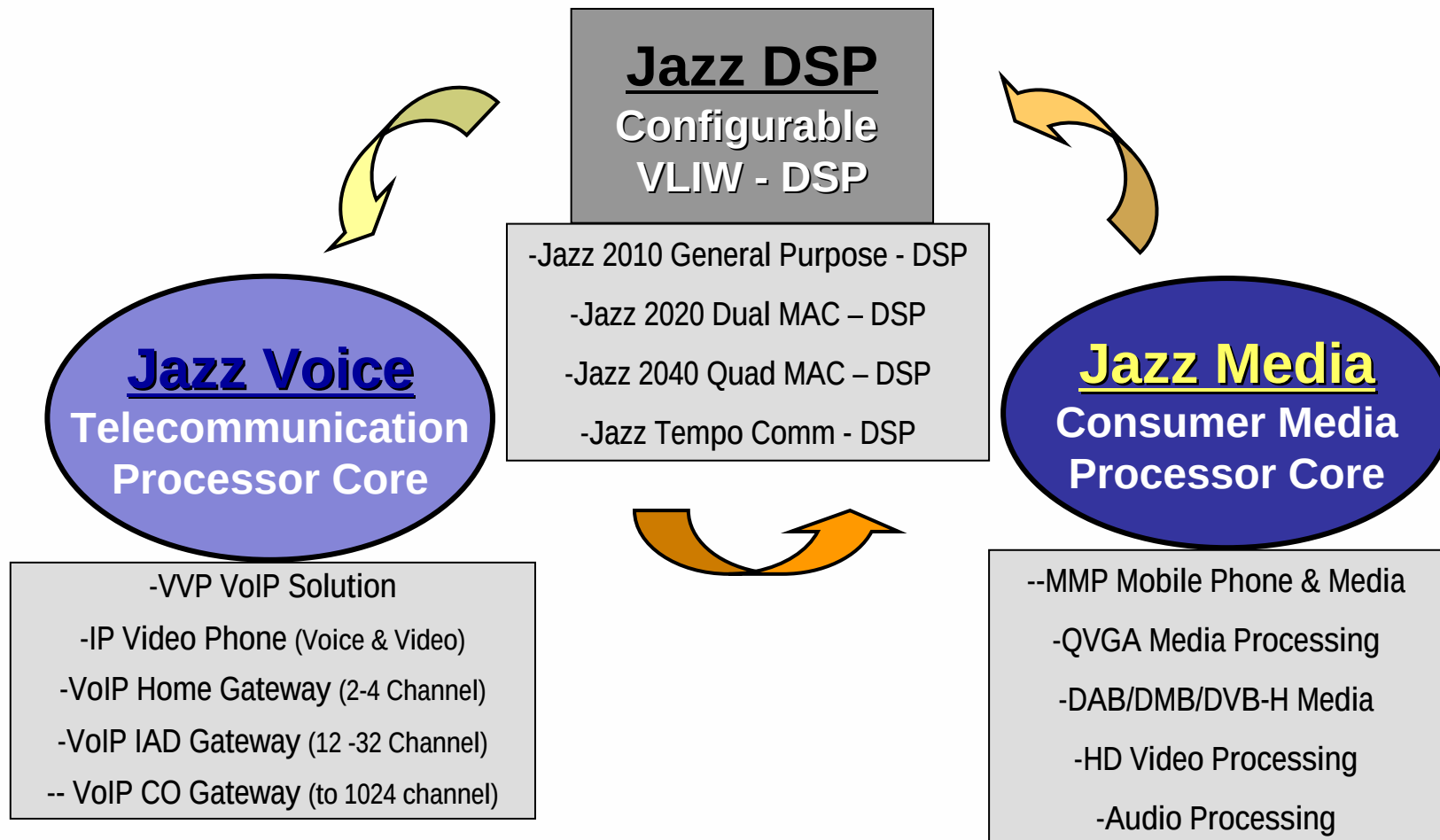
- Breakthrough configurable & pre-configured DSP architecture
- Industry's most efficient DSP and compiler combination*
- Industry's most versatile DSP to meet designer's application specific needs
 - Black-box Application Cores for Media Processing and VoIP
 - Highly Flexible Design methodology & development environment
 - Complete SoC integration and methodology
- Industry's only high level language based multi-processor DSP architecture
- Highest MIPs/mm2 – Lowest power/MIP
- Optimized application support for key DSP markets
- Proven technology

Jazz DSP is the first and only truly configurable DSP !

*Certified by EEMBC. Telemark score July 2002



Jazz Family - Product & Application Overview





Implementation Ready Pre-Configured Application Cores

Jazz Pre-Configured Application Processor Family

- Jazz Media - Mobile Media Processor (MMP)
- Jazz Media - Video Broadcast Media Processor (BMP)
- Jazz Voice - Home VoIP (VVP)
- Jazz Voice - IAD VoIP (Small Business)
- Jazz Audio (AAP)



Jazz Mobile Media Processor (MMP) Overview

The Jazz Media MMP cores can be used as a black box implementation of complete elementary stream processing for a mobile/portable media chip design. Each core can operate as a standalone block or as a slave to a host processor.

Each Jazz MMP core has a single, application-specific processor that combine a general-purpose instruction set with application-specific instructions that accelerate media processing functions.

The Jazz MMP processors are very-long instruction word (VLIW) processors containing embedded memory, a host bus interface (HBI) that can connect to most standard on-chip buses including AMBA, OCP and Wishbone, and a media-aware DMA memory interface called the Data Channel Interface (DCI).

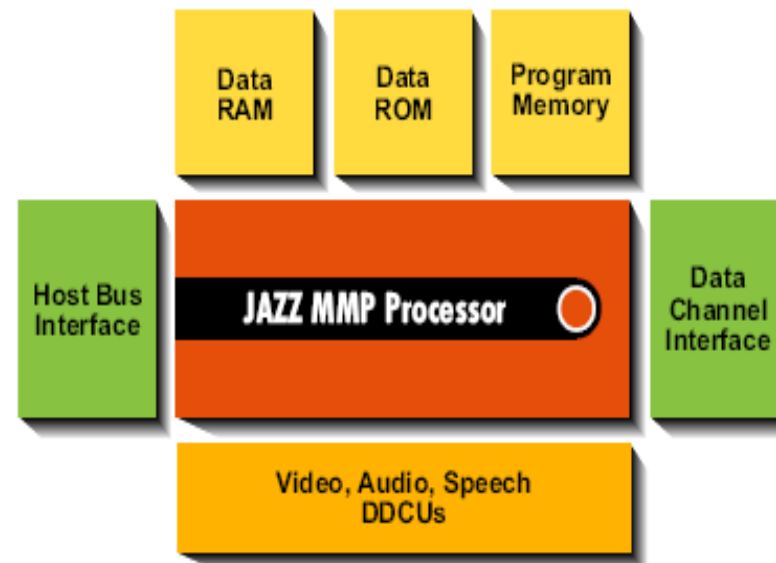
- *Complete silicon-ready solution*
- *Optimized for small die size and low power*
- *Low clock rates for advanced media standards (Available in audio, video and voice configurations)*
- *Reduces CPU processing load by 75-90%.*
- *Fully synthesizable and foundry independent*
- *Supported by a complete software development tool suite*
- *Optionally available as configurable cores for additional customization*



Jazz Mobile Media MMP - Implementation Ready Application Cores

The MMP media software modules have been designed to fully exploit the unique advantages of the MMP architecture. The MMP family includes the following modules:

- H.264 AVC Baseline Video Decoder
- MPEG4 Simple Profile Codec
- Window Media 9 Video Decoder
- JPEG Still Image Codec
- MPEG L1/L2 Audio Decoder
- MPEG L1/L2 Audio Encoder
- MP3 Audio Decoder
- MP3 Audio Encoder
- AAC / AAC+ Audio Decoder
- BSAC Audio Decoder
- Windows Media 9 Audio Decoder
- G.711, G.726, G.723.1, G.729a Codecs



All modules delivered in source code so you can add your own algorithms (e.g., motion estimation)



Jazz Mobile Media MMP - Implementation Ready Application Cores

The Jazz MMP family consists of the following specific products:

- **MMP Video Decoder Core** is optimized for video decoding and includes all MMP video/image decoding and image modules.
- **MMP Video Codec Core** is optimized for a full-duplex video codec and includes all MMP video/image decoding and codec modules.
- **MMP Media Decoder Core** is optimized for full media decoding and includes all media software modules.
- **MMP Media Codec Core** is optimized for a full-duplex media codec and includes all media software modules.

IMPLEMENTATION CHARACTERISTICS

	<u>GATE COUNT</u>	<u>CLOCK</u>	<u>POWER</u>
MMP Video Decoder	354K gates	50 MHz	23 mW
MMP Video Codec	367K gates	75 MHz	33 mW
MMP Media Decoder	386K gates	75 MHz	35 mW
MMP Media Codec	399K gates	125MHz	60 mW

These numbers represent the maximum requirements for running the MMP Media Software modules in a TSMC 0.13um process.



New Technology Convergence of Mobile Media Products



Mobile Integration
of VoIP & WiFi



S-DMB, DVB-H
& Satellite Radio
Integration



GPS Integration into
PDA/Smartphone



Integrated Micro
Projector with HDD



Mobile WiFi / Wi-Max
Steaming Media Device



Integrated DMB
Receiver for Live Digital TV



Jazz Mobile Media MMP - Platform & Application Support

Jazz MMP Processor

- 180, 130, 90nm process
- Ultra-low power performance
- Unique programmable, multiple engine VLIW architecture
- Supports ARM, MIPS & Open RISC
- Supports Linux 2.6, WinCE - OS
- Industry-leading graphics support with DSP-based video imaging, and audio accelerator
- Media processing for delivery and playback solution of native 3GPP and 3GPP2

Application Support

- Video broadcasting support for emerging standards like DVB-H, S-DMB, T-DMB and ISBD-T
- CIF, QVGA & D1 up to 720p resolution
- JPEG & M-JPEG imaging
- Advanced Audio - MP3, BSAC, AAC/AAC+, MPEG2 L1/L2 & Dolby®
- FPU support for 3D graphics & audio effects
- Microsoft (WMV), Apple (QuickTime) & Qualcomm (Brew-MediaFlo)

Jazz MMP - Simultaneous VLIW Parallel Processing



Jazz Media Processing - Hardware Accelerators

Jazz Media - H/W Acceleration – DDCUs for Media Processors

- BitStream
 - Reading and writing to/from bitstream buffer
- BitStream w/Huffman Tables
 - Same as above but includes loading and processing of Huffman tables for video; also reduces data memory requirements
- DCT/IDCT
 - Optimized datapath for DCT/IDCT operations
- IDCT
 - Optimized datapath for IDCT operations
- SIMD Adder
 - ALU optimized for video data widths and operations
- H-Interpolator
 - Interpolation and clipping operations
- MP3 Table Accelerator
 - Holds majority of MP3 Huffman tables and optimizes access; significantly reduces on-chip data memory requirements (over 3 sq. mm in 0.18).
- GMC
 - Optimizes key algorithms used in global motion compensation
- Six Tap
 - Six tap filter used in H.264 processing
- Inverse 4x4 Transform
 - Optimized for the H.264 4x4 transform
- CABAC /CAVLC
 - Acceleration of the CABAC/CAVLC processing in H.264
- ADLU
 - Speech processing accelerator

Custom H/W accelerators developed by Improv included within the Jazz Media Solution Kits:



Jazz Media Processing - Module Support

Jazz Media - Media Processing Codec's & Modules Overview

Video Support

- H.264 - Baseline, Main & High Profile Decoder
- MPEG4 - ASP Encoder
 - Full D1 Resolution
- MPEG4 - ASP Decoder
 - Broadcast Quality video at D1
 - Full ASP Implementation including ¼ pel, B frames, interlaced
- MPEG4 - SP Codec
 - Up to VGA sizes including error resiliency and data partitioning
- Windows Media 9 (WMV) Video Decoder
 - Simple and Main Profiles

Audio Support

- MPEG2 [MP@ML](#) Decoder
 - Fully compliant implementation
- AC-3 Dolby Digital™ & Prologic™ Decoder
 - Full surround sound or two channel sub-sampled
 - Dolby Licensed implementation 5.1 & 7.1
 - Only available to Dolby Licensees
- MP3 Audio – Encoder & Decoder
 - Complete implementation include 2.5 and LSR
- AAC & AAC+v2 Audio Decoder
- BSAC Audio Decoder
- Windows Media 9 (WMA) Audio Decoder

All modules delivered in source code so you can add your own algorithms (e.g., motion estimation)



Improv SOC Integration

- SOC Host Software Package
 - ANSI-C compliant API for configuring, controlling and running all application modules in an integrated framework
 - Supports up to 4 streams of data for video and audio
 - full on-chip debug (JTAG) integration
- Host Bus Interface – Jazz HBI
 - Hooks directly into host busses including AMBA™ AHB (ARM) and OCP
 - Can be configured for different custom busses and Interface
 - Mailbox-style interface for communicating with host processor
- Bit Stream FIFO
 - Efficient management and communication for loading the bit stream buffer
- Time Domain Manager
 - Time slice data interface
- Data Channel Interface – Jazz DCI
 - High-performance, “media aware” DMA for moving blocks of video data to and from memory
 - Different configurations for different memory layouts of video data in memory



Improv Tools & Support

The Jazz Tools Suites are required to develop and configure Jazz DSP cores.

The Jazz Standard Tool Suite is for software development and includes the Notation environment, Compiler/Assembler, Instruction-set Simulator and Profiler. The Jazz Composer Tool Suite includes all of the Standard Suite features and additionally allows customization of all Jazz DSP's and platforms.

- Tool Technology Summary

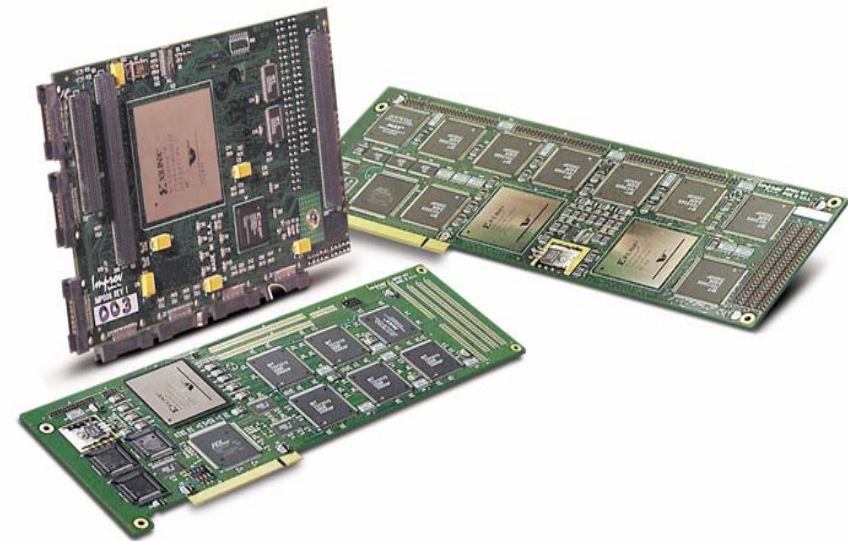
- Composer – graphical tool for creating custom platforms, processors and accelerators
- VLIW Compiler
- Profiling and Analysis Tools
- Java and C-based system-level design methodology
- Instruction-Set Simulator
- Automatic Generation of RTL databases and EDA scripts



Improv Emulation & Development Boards

Improv Jazz PSA Rehearsal™ Development Boards

- Processor and System Verification
 - Jazz PSA Rehearsal™ XE2-6K
*Dual FPGA (2-4 processor)
configuration; PCI-based*
 - Map “Composed” Jazz™ processors
onto FPGA technology, execute and test
application code - includes full debugger
support
 - Use same test bench for functional, ISS
and Rehearsal™ board simulation
 - Windows NT drivers – *Multiple
Configurations*





Processor Platform Offerings

Pre-Configured Application Processors

Improv 'Fixed' Core processors come pre-configured with "customer ready" Application Software, development tools and support greatly reducing internal engineering dependencies and providing fast project implementation for time to market constraints.

Includes:

1. Jazz DSP Platform
2. Application Software & Applicable Solution Kit
3. Improv Jazz Implementation Kit
4. Improv Jazz Standard Tool Suite

(includes complete software development environment, compiler/assembler, instruction-set simulator and profiler)

Configurable Processors

Improv "Configurable" DSP core processors offer versatility to meet designer's application specific needs when creating custom products and applications. Improv's Highly Flexible Design methodology & development Environment, complete SoC integration tools and the Industry's most efficient DSP and compiler combination provides an unparalleled advantage.

Includes:

1. Jazz DSP Platform
2. Application Software & Applicable Solution Kit
3. Improv Jazz Implementation Kit/Generator
4. Improv Jazz Composer Tool Suite

(includes all Standard Suite features, plus allows for complete customization of DSP & Application Platform)



Summary

- Improv's Configurable VLIW DSP processor architecture provides high performance processing engines and rapid design methodology to meet aggressive Time to Market requirements
- Flexible – programmable architecture meet the needs of broad application support and firmware upgradeability
- Complete Software Solution kits for vertical markets
- Integration and implementation tools and methodology
- Silicon implemented and customer proven technology